WASP 2026 · ASTROCLIMB SHARED TASK · IJCNLP–AACL 2026

Parameter-Efficient Adaptation of Qwen3-VL for Astronomy Citation Linking

Syed Mohaiminul Hoque | Center for Computational and Data Sciences Lab, Independent University, Bangladesh

FINAL TEST MACRO-F1

0.73725

+0.258 over best zero-shot VLM
on 2× T4 GPUs · 4-bit QLoRA

Code, notebooks & artifacts

TL;DR

We treat pair-relation classification as **one-token answer prediction** with Qwen3-VL, adapt it with **4-bit QLoRA on language attention**, average **three seeds**, and apply a free **modality mask**, reaching **0.73725** macro-F1. The hard part is **related_papers**.

1 The task

Given **two objects** from astronomy papers (each a **figure** or a **caption**), predict how their source papers are related. Labels are symmetric, exclusive, and defined by *hidden metadata* (DOI, citation edges).

0 same_figure — 10%
Figure + its own caption

1 same_paper — 30%
Different figures, same paper (DOI)

2 related_papers — 30%
Different papers, one cites the other

3 unrelated_papers — 30%
No relation

share of the 10K training labels

DATA & METRIC

10K	10K	3
labelled train pairs	test pairs (Kaggle)	pair types: CC · CI · II

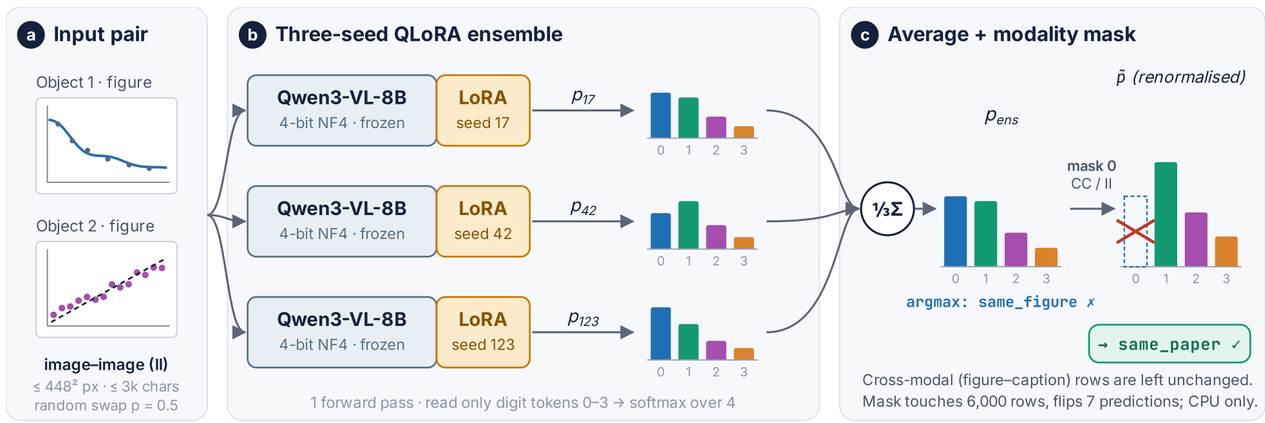
$F_{1,\text{macro}} = \frac{1}{4}\sum_c F_{1,c}$ — the rare `same_figure` class counts as much as the rest.

Controlled split: 200 pairs per class held out (seed 42) → 9.2K train / 800 val. No test metadata, DOI lookup or hidden labels used.

WHY IT IS HARD

- Inputs mix modalities: caption–caption, caption–image, image–image.
- same_paper vs. related_papers hinge on a shared DOI or a citation edge that never appears in the input.

2 System: constrained next-token classification



3 Training recipe

PROMPT AS CLASSIFIER

The system prompt defines the relations as digits **0–3**; objects appear in their native modality. Training: cross-entropy on the answer token. Inference reads only the four digit logits:

$$\hat{y} = \text{argmax}_{c\in\{0,1,2,3\}} z_{t_c}$$

QLORA

$$W_{\text{eff}} = Q_{\text{NF4}}(W_0) + (\alpha/r)\cdot BA, \quad r = 16,\ \alpha = 32$$

4-bit NF4 + double quant · LoRA on q/k/v/o (language) · lr 1e-4 · cosine · 1 epoch · global batch 16 · paged AdamW 8-bit · random object swap · images ≤ 448² · captions ≤ 3k chars

ENSEMBLE + MASK

$p_{\text{ens}} = \frac{1}{3}\sum_s p_s$; CC / II pairs: $\tilde{p}_0 = 0$, renormalise

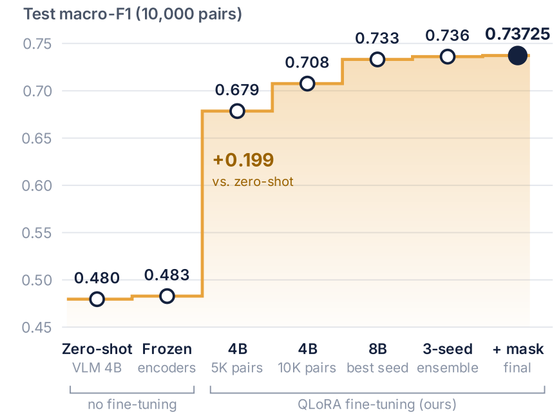
COST

15.3M	11.5 GiB	78 min
trainable params	peak GPU memory	inference, 10K pairs

WHAT DID *NOT* HELP

- ✗ Broader 4B adapters (visual / all linear): **+0.003–0.005**
- ✗ Restricted 4-way loss (8B): **0.73032**; blending it in: **−0.00014**
- ✗ Citation-graph auxiliary data (20K pairs): **−0.015** val

4 Results: the climb

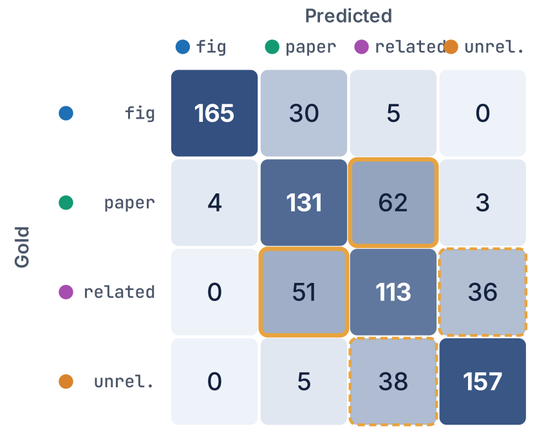


ZOOM: SEEDS → ENSEMBLE → MASK



Seeds: 0.73262 ± 0.00040. Test 95% half-width ≈ 0.008, so 4B→8B (+0.026) is resolvable and the ensemble gain indicative. The mask gain is certain: all 7 flipped rows were impossible `same_figure` predictions.

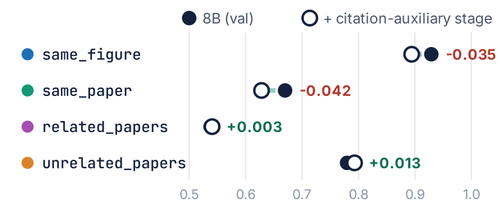
5 Where errors live



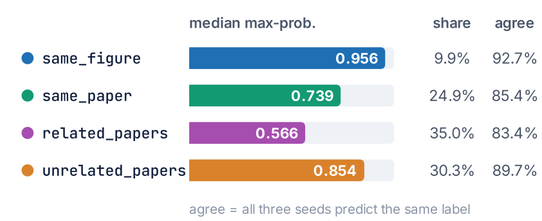
val. 800 pairs · 8B + citation-auxiliary model

187 of 234 errors: paper↔related (62/51, solid) and related↔unrelated (36/38, dashed). Symmetric → genuine ambiguity, not prior bias.

CITATION-AUXILIARY STAGE HURTS



TEST-SET CONFIDENCE, LABEL-FREE



agree = all three seeds predict the same label

Where the 3 seeds disagree (1,326 test rows = 13.3%)



1 = same_paper · 2 = related_papers · 3 = unrelated_papers · 0 = same_figure

FIGURES ARE HARDER THAN CAPTIONS



Takeaways: one-token QLoRA lifts Qwen3-VL from 0.48 → 0.73 · seeds + a task-valid mask add ≈ +0.004 for free · **related_papers** is the bottleneck, since citation links are often invisible in the pixels/text.

github.com/SyedT1/Astroclimb-sharedtask